What Your Agent Topology Is Actually Doing: Controlled Experiments on Communication Structure in Multi-Agent Systems

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Abstract

Multi-agent AI systems choose communication topologies—star, chain, ring, fully connected—with no controlled evidence for how topology affects behavior. We provide three lines of evidence. First, using island-model genetic algorithms as a model system (240 runs), we show topology independently explains 17% of behavioral diversity variance ($p < 10^{-7}$), amplified 14× on hard problems. Second, we validate the transfer to real agents: 3,200 LLM API calls across 8 coding tasks show that topology determines solution diversity in GPT-4o-mini agents. Under negative coupling ("be different"), the diversity ordering *inverts* from GAs (Cohen's $d = -11.1$); under positive coupling ("build on and improve"), it reverts to GA-like ordering ($\eta^2 = 0.291$, $p < 10^{-19}$). Changing one word in the prompt flips which topology produces the most diverse output. The sign depends on which adjoint of the topology the coupling mechanism instantiates—left Kan (homogenize) vs. right Kan (diversify). At least nine deployed systems chose topologies without controlled comparisons. Topology is a first-class safety variable; current systems are tuning it blindly.

1 Introduction

Every multi-agent system embeds a communication topology—round-robin group chat (AutoGen), DAG pipelines (LangGraph), manager hubs (CrewAI)—yet these choices are made without controlled evidence. Dochkina [Dochkina, 2026] ran 25,000 tasks across 256 agents, finding a 44% quality spread ($d = 1.86$) between topologies. Hierarchical deployments fail 36% of the time; swarms fail 68%. Topology is doing something—but *what, exactly?*

We provide the first controlled answer, bridging from a model system (genetic algorithms) to real LLM agents. Our contributions: (1) topology independently determines behavioral diversity in both GAs and LLM agents; (2) the effect *inverts* between GAs and agents, explained by which adjoint of the topology the coupling mechanism instantiates; (3) at least nine deployed systems chose topologies without controlled comparisons.

2 Island Models as Agent Systems

An island-model GA partitions a population across sub-populations (*islands*) that evolve independently, exchanging individuals periodically via *migration* channels. The mapping to agent orchestration is direct:

GA concept	Agent equivalent
Island	Agent (local state, computation)
Migration channel	Message channel
Population diversity	Solution diversity
Fitness function	Task performance metric
Migration topology	Orchestration graph

The analogy is structural, not metaphorical: both systems have nodes with local state exchanging information along directed channels. The agent-side quantities are measured directly: **agent diversity** = mean pairwise cosine distance on sentence embeddings (all-MiniLM-L6-v2) of agent outputs; **agent fitness** = task performance (test pass rate, correctness). GAs provide cheap experimental control (hundreds of trials); the LLM experiment (Section 4.3) validates transfer to real agents.

3 Definitions

Definition 1 (Communication topology)**.** *A* communication topology *for n agents is a directed graph $D = (V, A)$ with $|V| = n$, where each arc $(u, v) \in A$ is a permitted channel from agent u to agent v.*

Definition 2 (Cycle rank)**.** *The* cycle rank $\beta_1(G) = |E| - |V| + 1$ *(for connected G) counts independent*

cycles—feedback loops in the communication structure.

Definition 3 (Directed cycle count)**.** *$\kappa(D)$ counts distinct simple directed cycles in D [Johnson, 1975]. Unlike β_1, this depends on arc* arrangement*, not just edge count. A DAG has $\kappa = 0$.*

NK landscapes [Kauffman, 1993] are tunably rugged fitness functions over binary strings of length N, where each bit's contribution depends on K other bits. K=0 gives a smooth landscape; increasing K adds local optima.

4 Experimental Evidence

We present three experiments spanning GA model systems and LLM agents, totaling 3,600+ runs and API calls.

4.1 Experiment 1: Topology affects diversity at constant density

Comparing a chain to a complete graph conflates density and cycle structure: $\beta_1 = m - n + 1$ [Hatcher, 2002] means cycle count and edge count are perfectly correlated for undirected graphs. We escape via directed cycle count κ, which varies independently of density.

Design. 8 directed graphs, all n=8, 16 arcs (δ=2.0), varying only in $\kappa \in \{0, 0, 3, 10, 14, 20, 29, 47\}$. Island-model GA: 50 individuals/island, 500 generations, 30 seeds per topology, 240 runs total.

Results. $F(7, 232) = 6.90$, $p = 1.85 \times 10^{-7}$, $\eta^2 = 0.17$. More directed cycles *accelerate consensus* ($r = -0.68$): topologies with more feedback loops cause agents to converge faster, reducing capacity to maintain alternative strategies.

4.2 Experiment 2: Hard problems amplify the effect 14×

On NK landscapes with tunable ruggedness, topology's effect scales with difficulty:

K	η^2 (diversity)	Interpretation
0 (trivial)	0.05	Topology barely matters
2 (moderate)	0.20	Moderate effect
4 (hard)	0.45	Large effect
6 (very hard)	0.69	Topology dominates

Simple tasks are structurally smooth—topology barely matters. Complex tasks (multiple valid strategies, ambiguous evaluation) are rugged—topology dominates. Real-world deployments overwhelmingly involve the latter.

4.3 Experiment 3: Topology inverts diversity in LLM agents

The GA experiments establish that topology controls diversity. But does this transfer to LLM agents? We test directly.

Design. 5 GPT-4o-mini agents, 8 coding tasks (shortest path, kth largest, rate limiter, JSON parser, LRU cache, regex matcher, expression evaluator, Bloom filter), 4 topologies (none, ring, star, fully connected), 10 seeds per condition. 1,600 API calls total. Coupling is *negative*: each agent is instructed to use a *different* algorithm from those visible in its context. Diversity is measured as mean pairwise cosine distance on sentence embeddings (all-MiniLM-L6-v2) of agent outputs.

Results. The diversity ordering **inverts** from the GA result:

System	Diversity ordering	Statistic
GA	none > ring > FC	$W = 1.0$ (6 domains)
LLM	FC > ring ≈ star > none	$W = 0.91$ (6/8 tasks)

Six of eight tasks show the inverted ordering cleanly (Kruskal-Wallis $p < 10^{-6}$; Cohen's d ranges from -14.8 to -1.9). Two exceptions are informative: bloom filter ($d \approx 0$, unimodal task—no algorithmic variety for topology to act on) and expression evaluator ($d = +0.4$, the model's prior is already spread).

Why the sign flips. Under "none" topology, all 5 LLM agents independently generate the *same* algorithm (e.g., quickselect for kth_largest). LLMs start from a **peaked prior** (low entropy); GAs start from **random initialization** (high entropy). Isolation preserves initial entropy: high for GAs (diverse), low for LLMs (homogeneous). The inversion follows from which *adjoint* of the topology the coupling mechanism instantiates:

- **GA migration** = **left Kan extension** (colimit-like): merges populations. More connectivity → less diversity.

Table 1: Agent systems whose topology choices lack controlled evidence.

System	Issue
Graph-GRPO [Cang et al., 2026]	Complete = worst; also max β_1
OFA-MAS [Li et al., 2026]	Prunes density & cycles together
G-Designer [Zhang et al., 2025]	No density-controlled baseline
GoAgent [Chen et al., 2026]	DAG vs. dense alternatives
ARG-Designer [Li et al., 2026b]	DAG space only
AgentConductor [Wang et al., 2026]	Cycles never tested
DyTopo [Lu et al., 2026]	DAG enforced throughout
HERA [Li and Ramakrishnan, 2026]	Cycles tracked, not controlled
Dochkina [Dochkina, 2026]	Density & cycles confounded

- **LLM "be different" context = right Kan extension** (limit-like): accumulates exclusion constraints. More connectivity → agents forced into distributional tails → *more* diversity.

The topology is the invariant; the sign depends on the adjoint. $\mathrm{Lan}_F \dashv \mathrm{Ran}_F$ reverses ordering on any poset, including the diversity ordering on topologies.

Positive coupling confirms the prediction. We repeated the experiment with *positive* coupling: each agent is instructed to "build on and improve" neighbors' solutions instead of "be different." 320 additional API calls (total: 3,200). Results: the diversity ordering **reverts to GA-like**: none > star ≈ ring > FC (Kruskal-Wallis $H = 94.83$, $p = 2.01 \times 10^{-20}$; Cohen's $d = 1.08$, none vs. FC; $\eta^2 = 0.291$). Crucially, the *none* topology—where agents have no neighbors—shows identical diversity across conditions (negative: 0.090, positive: 0.095, $p = 0.96$), confirming that coupling sign, not some other experimental variable, drives the reversal. Connected topologies collapse to near-zero diversity (0.025–0.028) under positive coupling, as agents converge on a shared best solution. One word change in the prompt flips the diversity ordering—precisely as the Lan ⊣ Ran framework predicts.

5 Implications for Agent Orchestration

The evidence gap. At least nine recent agent systems—spanning ICML, AAAI, WWW, and NeurIPS—chose topologies without density-controlled comparisons (Table 1).

Safety risks. Our LLM experiment sharpens three concerns: *(1) The sign matters.* Changing one word in the coupling instruction flips which topology produces the most diverse output ($\eta^2 = 0.291$, $p < 10^{-19}$). Predictions based on GA intuition alone will have the wrong sign for diversifying agents. *(2) Premature consensus.* Isolated LLM agents converge to identical solutions (Cohen's $d = -11.1$ vs. connected agents). "Ensemble diversity" claims based on independent agents may be illusory when models share a peaked prior. *(3) Error amplification.* Moran [Moran, 2025] reports 17× error amplification in unstructured communication. The standard fix (impose DAG structure) simultaneously removes cycles *and* density—we cannot know which helps.

Recommendations.

1. **Treat topology as a first-class variable.** It controls the diversity–convergence tradeoff—in both GAs and LLM agents, empirically confirmed.
2. **Know your coupling sign.** One word change in the coupling instruction flips the diversity ordering empirically. Building on neighbors' outputs (left Kan) homogenizes; differentiating from them (right Kan) diversifies.
3. **Match topology to task complexity.** Simple tasks tolerate any topology; hard tasks demand deliberate choice (η^2 scales 14× from K=0 to K=6).
4. **Monitor diversity, not just accuracy.** Topology effects manifest in strategy diversity. Accuracy alone masks topology-induced failures.

6 Related Work

Yang et al. [Yang et al., 2025] identify topology as the underexplored MAS dimension; we provide the controlled experiments they call for. Dochkina [Dochkina, 2026] shows massive effects (25K tasks, $d = 1.86$) without isolating the mechanism. HERA [Li and Ramakrishnan, 2026] tracks cycle count during evolution but does not control density. In evolutionary computation, the topology–diversity link is established [Tomassini, 2005, Skolicki and De Jong, 2005] without density control. Capucci and Myers [Capucci and Myers, 2026] formalize the cost: each cycle requires an independent convergence guarantee.

7 Conclusion

Communication topology is not neutral infrastructure—it is a design variable controlling how multi-agent systems balance exploration against convergence. In GA model systems, topology independently explains 17% of diversity variance on easy problems, rising to 69% on hard ones. In

LLM agents, the effect is equally strong but its *sign* depends on coupling: negative coupling ("be different") inverts the GA ordering ($d = -11.1$); positive coupling ("build on") restores it ($\eta^2 = 0.291$). The sign is determined by which adjoint the coupling instantiates—left Kan (homogenize) vs. right Kan (diversify)—confirmed by the null control: isolated agents show identical diversity regardless of coupling instruction ($p = 0.96$). At least nine deployed systems chose topologies based on comparisons that simultaneously varied the property they claimed to test. For safe multi-agent deployment, topology must be tuned deliberately, its coupling sign understood, and its effects monitored through diversity metrics—not assumed away.

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